

I. Normal Cases

Sequential Access (LRU + MRU)

Read Policy: Load-through

MACHINE 74-WAY BSA - LRU VS MRU

Run to end?

BLOCK SIZE (WORDS)16

CACHE BLOCKS16

READ POLICYLoad-thruNon-load

TEST CASE(a) Sequential

VIEW MODEStepSnapshot

4 sets 4 ways 64 accesses

FINAL STATE64 / 64

MEMORY BLOCK31

SETs3

TAG0x7

LRU MISS - evict 0x3

MRU HIT

CACHE ARRAY - SETS = 4 WAYS

Hit Miss Evicted Valid Empty

U1 - LRU0 hit 64 miss

	WAY 0	WAY 1	WAY 2	WAY 3
S0	0x4 blk 16	0x5 blk 20	0x6 blk 24	0x7 blk 28
S1	0x4 blk 17	0x5 blk 21	0x6 blk 25	0x7 blk 29
S2	0x4 blk 18	0x5 blk 22	0x6 blk 26	0x7 blk 30
S3	0x4 blk 19	0x5 blk 23	0x6 blk 27	0x7 blk 31 EVICT

U2 - MRU16 hit 48 miss

	WAY 0	WAY 1	WAY 2	WAY 3
S0	0x0 blk 0	0x1 blk 4	0x6 blk 24	0x7 blk 28
S1	0x0 blk 1	0x1 blk 5	0x6 blk 25	0x7 blk 29
S2	0x0 blk 2	0x1 blk 6	0x6 blk 26	0x7 blk 30
S3	0x0 blk 3	0x1 blk 7	0x6 blk 27	0x7 blk 31 HIT

Stats - LRU vs MRU

	LRU	MRU
Total accesses	64	64
cache hits	0	16
cache misses	64	48
Hit rate	0.0%	25.0%
Miss rate	100.0%	75.0%
AMAT (cycles)	86	64.75
Total access time	704	784

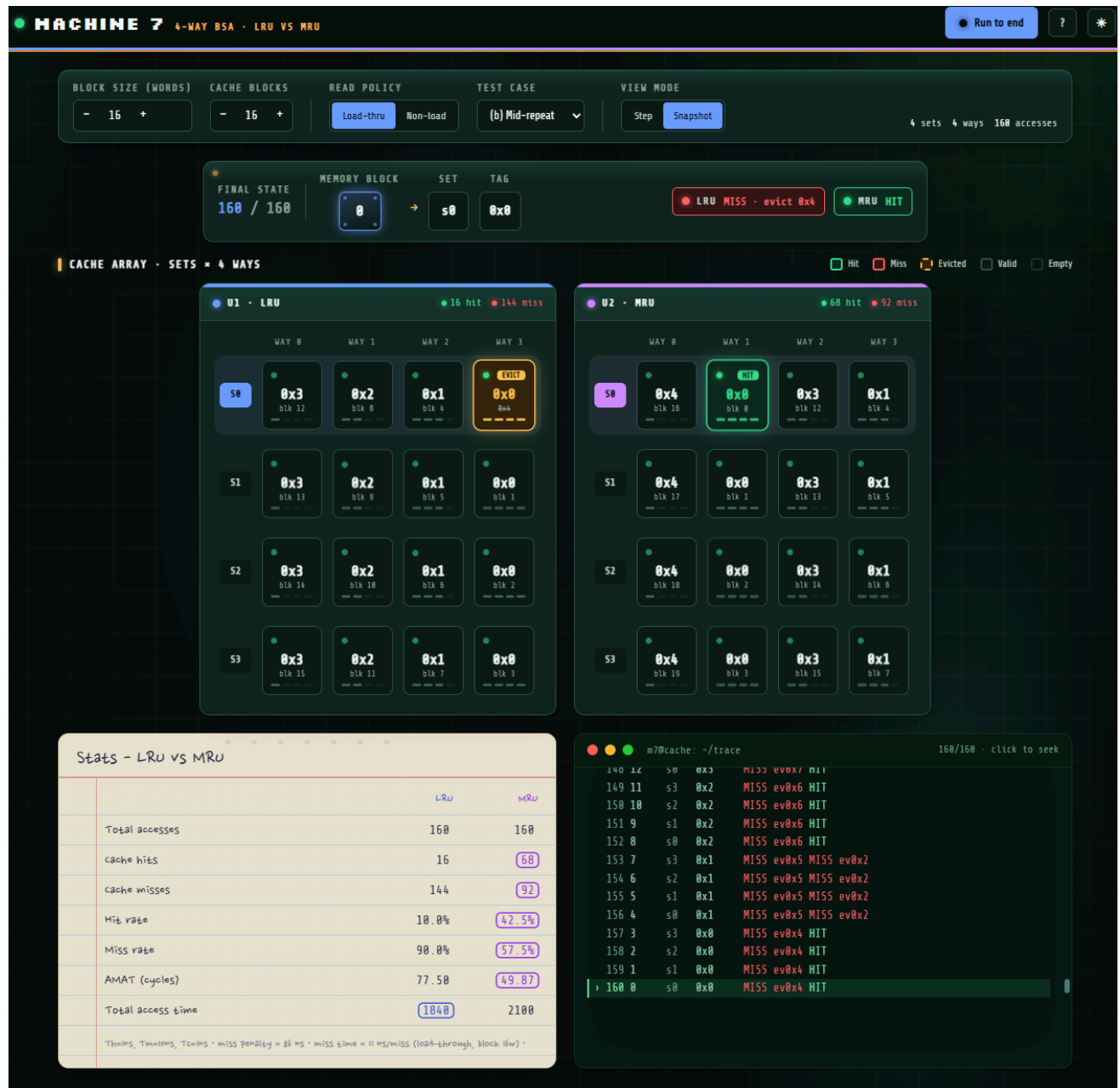
Thrains, Tmaloens, Tcains - Miss Penalty = 86 ns - Miss time = 11 ns/miss (load-through, block idw) -

m7@cache: ~/trace

64/64 - click to seek

```
54 21 s1 0x5 MISS ev0x1 MISS ev0x4
55 22 s2 0x5 MISS ev0x1 MISS ev0x4
56 23 s3 0x5 MISS ev0x1 MISS ev0x4
57 24 s0 0x6 MISS ev0x2 MISS ev0x5
58 25 s1 0x6 MISS ev0x2 MISS ev0x5
59 26 s2 0x6 MISS ev0x2 MISS ev0x5
60 27 s3 0x6 MISS ev0x2 MISS ev0x5
61 28 s0 0x7 MISS ev0x3 HIT
62 29 s1 0x7 MISS ev0x3 HIT
63 30 s2 0x7 MISS ev0x3 HIT
64 31 s3 0x7 MISS ev0x3 HIT
```

Read Policy: Load-through



Random Sequence (LRU + MRU)
Read Policy: Load-through

MACHINE 74-WAY BSA • LRU VS MRU

Run to end

BLOCK SIZE (WORDS)

-16+

CACHE BLOCKS

-16+

READ POLICY

Load-thru

Non-load

TEST CASE

(c) Random (64)

Regenerate

VIEW MODE

Step

Snapshot

4 sets 4 ways 64 accesses

FINAL STATE

64 / 64

MEMORY BLOCK

40

SET

s0

TAG

0xA

LRU MISS - evict 0x27

MRU MISS - evict 0xEF

CACHE ARRAY • SETS = 4 WAYS

Hit

Miss

Evicted

Valid

Empty

U1 • LRU

1 hit 63 miss

WAY 0

WAY 1

WAY 2

WAY 3

S0

0xEB

blk 940

0xEF

blk 956

0xA

blk 244

0x3D

blk 244

S1

0xB1

blk 789

0x1D

blk 117

0xAD

blk 893

0xEC

blk 945

S2

0xDD

blk 886

0xD5

blk 854

0x93

blk 598

0x28

blk 138

S3

0x43

blk 271

0xB7

blk 735

0x66

blk 411

0xF3

blk 975

U2 • MRU

2 hit 62 miss

WAY 0

WAY 1

WAY 2

WAY 3

S0

0x87

blk 540

0xFB

blk 1884

0x6D

blk 436

0xA

blk 66

S1

0x67

blk 413

0x3F

blk 253

0xC4

blk 785

0x1D

blk 117

S2

0xA0

blk 842

0x0

blk 2

0xD5

blk 854

0x34

blk 218

S3

0xF7

blk 991

0x47

blk 287

0x9C

blk 627

0xF3

blk 975

Stats - LRU VS MRU

	LRU	MRU
Total accesses	64	64
cache hits	1	2
cache misses	63	62
Hit rate	1.6%	3.1%
Miss rate	98.4%	96.9%
AMAT (cycles)	84.67	83.34
Total access time	709	714

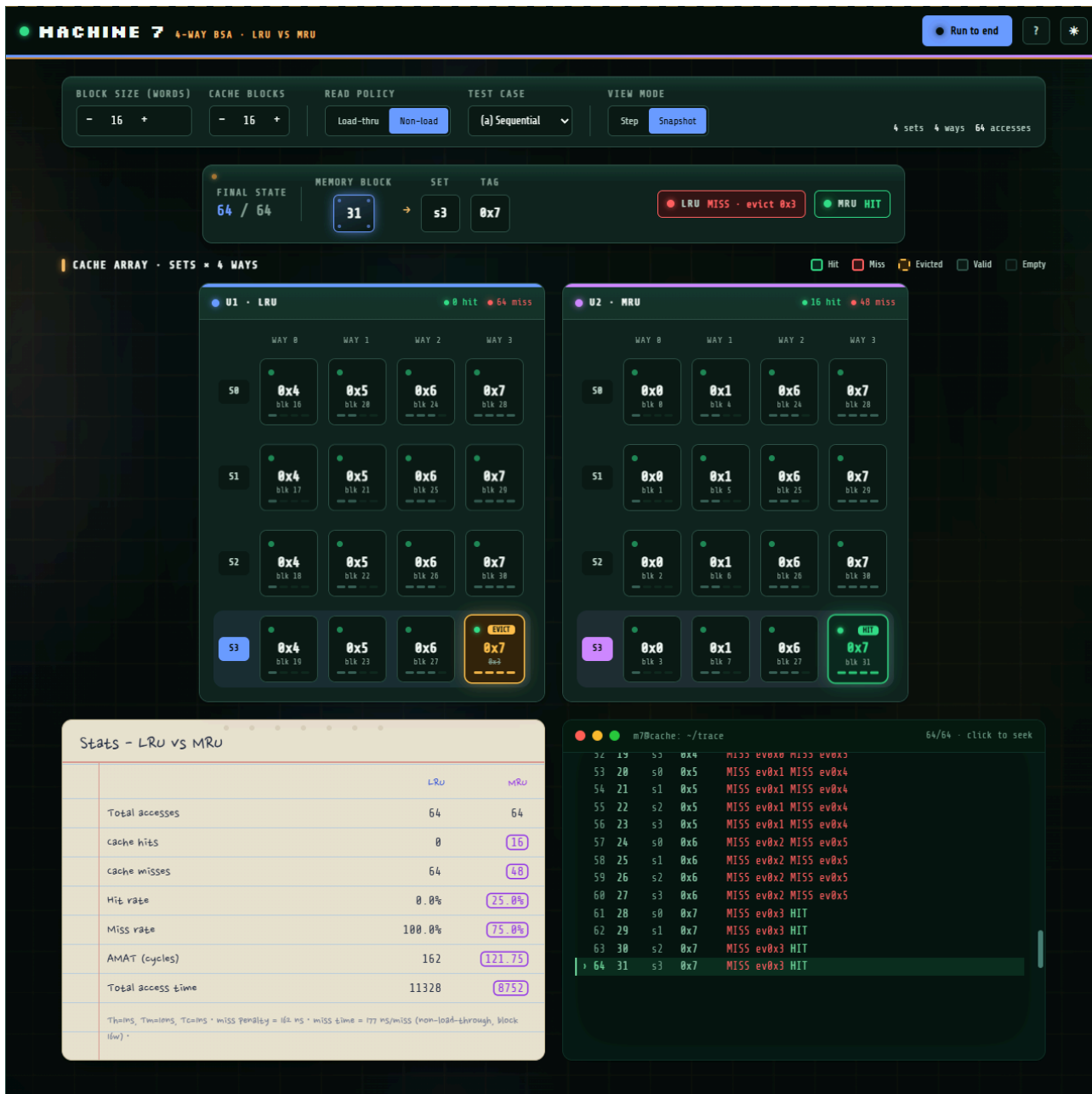
Th=1ns, Tm=10ns, Tc=1ns • miss penalty = 86 ns • miss time = 11 ns/miss (load-through, block 16w) •

m7@cache: ~/trace

64/64 - click to seek

```
54 411 s3 0x66 MISS ev0x26MISS ev0xB7
55 389 s1 0x4D MISS ev0x30MISS ev0xFD
56 693 s1 0xA0 MISS ev0xC MISS ev0xFD
57 975 s3 0xF3 MISS ev0xB9MISS ev0x66
58 945 s1 0xEC MISS ev0x9FMISS ev0xAD
59 789 s1 0xB1 MISS ev0xF0MISS ev0xEC
60 940 s0 0xEB MISS ev0xC2MISS ev0x3D
61 854 s2 0xD5 MISS ev0xFEMISS ev0xDD
62 956 s0 0xEF MISS ev0xC5MISS ev0xEB
63 117 s1 0x1D MISS ev0x40MISS ev0xB1
64 40 s0 0xA MISS ev0x27MISS ev0xEF
$
```

Sequential Access (LRU + MRU)
Read Policy: Non Load-through



Mid-repeat Blocks (LRU + MRU)
Read Policy: Non Load-through

MACHINE 7

4-WAY BSA • LRU VS MRU

Run to end

?

*

BLOCK SIZE (WORDS)

- 16 +

CACHE BLOCKS

- 16 +

READ POLICY

Load-thru

Non-load

TEST CASE

(b) Mid-repeat

VIEW MODE

Step

Snapshot

4 sets

4 ways

160 accesses

FINAL STATE

160 / 160

MEMORY BLOCK

0

SET

s0

TAG

0x0

LRU MISS • evict 0x4

MRU HIT

CACHE ARRAY • SETS = 4 WAYS

Hit

Miss

Evicted

Valid

Empty

U1 • LRU

16 hit • 144 miss

WAY 0

WAY 1

WAY 2

WAY 3

50

0x3

blk 12

0x2

blk 0

0x1

blk 4

0x0

blk 8

51

0x3

blk 13

0x2

blk 0

0x1

blk 5

0x0

blk 1

52

0x3

blk 14

0x2

blk 10

0x1

blk 6

0x0

blk 2

53

0x3

blk 15

0x2

blk 11

0x1

blk 7

0x0

blk 3

U2 • MRU

68 hit • 92 miss

WAY 0

WAY 1

WAY 2

WAY 3

50

0x4

blk 16

0x0

blk 0

0x3

blk 12

0x1

blk 4

51

0x4

blk 17

0x0

blk 1

0x3

blk 13

0x1

blk 5

52

0x4

blk 18

0x0

blk 2

0x3

blk 14

0x1

blk 6

53

0x4

blk 19

0x0

blk 3

0x3

blk 15

0x1

blk 7

Stats - LRU VS MRU

	LRU	MRU
Total accesses	160	160
cache hits	16	68
cache misses	144	92
Hit rate	10.0%	42.5%
Miss rate	90.0%	57.5%
AMAT (cycles)	145.90	93.57
Total access time	25744	17372

Things: Timings, Tcoins • miss penalty = 16x ms • miss time = 777 ms/miss (non-load-through, block (6w)) •

m7@cache: ~/trace

160/160 - click to seek

140 12 s0 0x2 MISS ev0x7 HIT

149 11 s3 0x2 MISS ev0x6 HIT

150 10 s2 0x2 MISS ev0x6 HIT

151 9 s1 0x2 MISS ev0x6 HIT

152 8 s0 0x2 MISS ev0x6 HIT

153 7 s3 0x1 MISS ev0x5 MISS ev0x2

154 6 s2 0x1 MISS ev0x5 MISS ev0x2

155 5 s1 0x1 MISS ev0x5 MISS ev0x2

156 4 s0 0x1 MISS ev0x5 MISS ev0x2

157 3 s3 0x0 MISS ev0x4 HIT

158 2 s2 0x0 MISS ev0x4 HIT

159 1 s1 0x0 MISS ev0x4 HIT

160 0 s0 0x0 MISS ev0x4 HIT

Random Sequence (LRU + MRU)
Read Policy: Non Load-through

● MACHINE 7

4-WAY BSA • LRU VS MRU

● Run to end

?

✱

BLOCK SIZE (WORDS)

- 16 +

CACHE BLOCKS

- 16 +

READ POLICY

Load-thru

Non-load

TEST CASE

(c) Random (64)

Regenerate

VIEW MODE

Step

Snapshot

4 sets

4 ways

64 accesses

FINAL STATE

64 / 64

MEMORY BLOCK

40

→

SET

s0

TAG

0xA

LRU MISS - evlct 0x27

MRU MISS - evlct 0xEF

CACHE ARRAY • SETS = 4 WAYS

Hit

Miss

Evicted

Valid

Empty

U1 • LRU

1 hit 63 miss

WAY 0

WAY 1

WAY 2

WAY 3

50

0xEB

blk 948

0xEF

blk 956

0xA

0x27

0x3D

blk 244

51

0xB1

blk 709

0x10

blk 117

0xAD

blk 693

0xEC

blk 945

52

0xDD

blk 806

0xD5

blk 854

0x93

blk 508

0x20

blk 110

53

0x43

blk 271

0xB7

blk 735

0x66

blk 431

0xF3

blk 975

U2 • MRU

2 hit 62 miss

WAY 0

WAY 1

WAY 2

WAY 3

50

0x07

blk 548

0xFB

blk 1004

0x6D

blk 430

0xA

0xEF

blk 210

51

0x67

blk 433

0x3F

blk 253

0xC4

blk 785

0x1D

blk 117

52

0xA0

blk 842

0x0

blk 2

0xD5

blk 854

0x34

blk 210

53

0xF7

blk 891

0x47

blk 287

0x9C

blk 627

0xF3

blk 975

Stats - LRU vs MRU

	LRU	MRU
Total accesses	64	64
cache hits	1	2
cache misses	63	62
Hit rate	1.6%	3.1%
Miss rate	98.4%	96.9%
AMAT (cycles)	159.48	156.97
Total access time	11167	11006
*Htms, Thtms, Tctms * MISS penalty = 164 ns * MISS time = 777 ns/MISS (non-load-through, block flow) *		

m7@cache: ~/trace

64/64 • click to seek

54 411 s3 0x66 MISS ev0x26MISS ev0x07

55 309 s1 0x40 MISS ev0x30MISS ev0xF0

56 693 s1 0xAD MISS ev0xC MISS ev0x40

57 975 s3 0xF3 MISS ev0xB9MISS ev0x66

58 945 s1 0xEC MISS ev0x9FMISS ev0xAD

59 709 s1 0xB1 MISS ev0xF0MISS ev0xEC

60 940 s0 0xEB MISS ev0xC2MISS ev0x3D

61 854 s2 0xD5 MISS ev0xFEMISS ev0xD0

62 956 s0 0xEF MISS ev0xC5MISS ev0xEB

63 117 s1 0x10 MISS ev0x40MISS ev0xB1

64 40 s0 0xA MISS ev0x27MISS ev0xEF

\$

II. Special Cases

Maximum Set (LRU + MRU)

Inputs: 0, 4, 8, 12, 16, 0, 4, 8, 12, 16, 0, 4, 8, 12, 16

Read Policy: Load-through

MACHINE 7

4-WAY BSA · LRU VS MRU

Run to end ? *

BLOCK SIZE (WORDS)

CACHE BLOCKS

READ POLICY

TEST CASE

VIEW MODE

- 16 +

- 16 +

Load-thru Non-load Custom ▾

Step Snapshot

4 sets 4 ways 15 accesses

CUSTOM SEQUENCE - COMMA/SPACE SEPARATED BLOCK NUMBERS {0-1023}

0, 4, 8, 12, 16, 0, 4, 8, 12, 16, 0, 4, 8, 12, 16

FINAL STATE
15 / 15

MEMORY BLOCK
16

→ SET
s0

TAG
0x4

LRU MISS · evict 0x0

MRU HIT

CACHE ARRAY · SETS × 4 WAYS

☒ Hit
 ☐ Miss
 ☐ Evicted
 ☐ Valid
 ☐ Empty

U1 · LRU 8 hit 15 miss

	WAY 0	WAY 1	WAY 2	WAY 3
S0	0x2 blk 0	0x3 blk 12	0x4 evict 0x0	0x1 blk 4
S1	.	.	.	.
S2	.	.	.	.
S3	.	.	.	.

U2 · MRU 8 hit 7 miss

	WAY 0	WAY 1	WAY 2	WAY 3
S0	0x0 blk 0	0x2 blk 8	0x3 blk 12	0x4 hit blk 16
S1	.	.	.	.
S2	.	.	.	.
S3	.	.	.	.

Stats - LRU vs MRU

	LRU	MRU
Total accesses	15	15
cache hits	0	8
cache misses	15	7
Hit rate	0.0%	53.3%
Miss rate	100.0%	46.7%
AMAT (cycles)	86	40.67
Total access time	165	285

Timing: Timelms, Tcmms = miss penalty = 16 ns · miss time = 11 ns/misss (log6-through, block 6w) ·

```
m7@cache: ~/trace 15/15 - click to seek
5 16 s0 0x4 MISS ev0x0 MISS ev0x3
6 0 s0 0x0 MISS ev0x1 HIT
7 4 s0 0x1 MISS ev0x2 HIT
8 8 s0 0x2 MISS ev0x3 HIT
9 12 s0 0x3 MISS ev0x4 MISS ev0x2
10 16 s0 0x4 MISS ev0x0 HIT
11 0 s0 0x0 MISS ev0x1 HIT
12 4 s0 0x1 MISS ev0x2 HIT
13 8 s0 0x2 MISS ev0x3 MISS ev0x1
14 12 s0 0x3 MISS ev0x4 HIT
15 16 s0 0x4 MISS ev0x0 HIT
$
```

Read Policy: Load-through

	LRU	MRU
Total accesses	10	10
cache hits	9	9
cache misses	1	1
Hit rate	90.0%	90.0%
Miss rate	10.0%	10.0%
AMAT (cycles)	9.50	9.50
Total access time	155	155

$\text{Thrashing} = \text{Thrashers} \cdot \text{TCRs} \cdot \text{miss penalty} = 20 \text{ ms} \cdot \text{miss time} = 11 \text{ ms/miss}$ (load-through, block (lw) -

Read policy: Load-through

	LRU	MRU
Total accesses	16	16
cache hits	0	0
cache misses	16	16
Hit rate	0.0%	0.0%
Miss rate	100.0%	100.0%
AMAT (cycles)	86	86
Total access time	176	176

$T_{\text{hits}} = T_{\text{misses}} = T_{\text{cache}} = 11 \text{ ns}$
 $\text{miss penalty} = 86 \text{ ns}$
 $\text{miss time} = 11 \text{ ns/miss (load-through, block fwd.)}$

III. Edge Cases

Lowest and Highest legal block indices

Input: 0, 1023, 0, 1023

Read policy: Load-through

The screenshot displays the m7Bcache simulator interface. At the top, the title bar reads "MACHINE 7 4-WAY BSA - LRU VS MRU". The main control panel includes fields for "BLOCK SIZE (WORDS)" (16), "CACHE BLOCKS" (16), "READ POLICY" (Load-thru), "TEST CASE" (Custom), and "VIEW MODE" (Step/Snapshot). The "CUSTOM SEQUENCE" field contains "0, 1023, 0, 1023". Below this, the "FINAL STATE" shows "4 / 4" blocks, with a "MEMORY BLOCK" of 1023, "SET" s3, and "TAG" 0xFF. The "CACHE ARRAY" section shows two cache views: "U1 - LRU" and "U2 - MRU". Both views show a 4x4 grid of cache blocks. In the LRU view, the first three rows are empty, and the fourth row contains block 0xFF (tag 1023) in set s3. In the MRU view, the first three rows are empty, and the fourth row contains block 0xFF (tag 1023) in set s3. A legend indicates Hit (green), Miss (red), Evicted (orange), Valid (blue), and Empty (grey). At the bottom left, a table titled "Stats - LRU vs MRU" compares the two policies. At the bottom right, a terminal window shows the command "run trace --policy lru,mru" and its output.

	LRU	MRU
Total accesses	4	4
cache hits	2	2
cache misses	2	2
Hit rate	50.0%	50.0%
Miss rate	50.0%	50.0%
AMAT (cycles)	43.50	43.50
Total access time	54	54
Times, TimesMS, TimesS = miss penalty = 86 ns * miss time = 11 ns/miss (load-through, block 0w) *		

```
$ run trace --policy lru,mru
# blk set tag lru mr
1 0 s0 0x0 MISS MISS
2 1023 s3 0xFF MISS MISS
3 0 s0 0x0 HIT HIT
4 1023 s3 0xFF HIT HIT
$
```

Out of Range inputs
Input: 0, 1, 1024
Read Policy: Load-through

BLOCK SIZE (WORDS)

- 16 +

CACHE BLOCKS

- 16 +

READ POLICY

Load-thru

Non-load

TEST CASE

Custom

VIEW MODE

Step

Snapshot

4 sets 4 ways 2 accesses

CUSTOM SEQUENCE - COMMA/SPACE SEPARATED BLOCK NUMBERS (0-1023)

0, 1, 1024

1024 is out of range (0..1023)

FINAL STATE

2 / 2

MEMORY BLOCK

1

SET

s1

TAG

0x0

LRU MISS

MRU MISS

CACHE ARRAY - SETS x 4 WAYS

U1 - LRU

0 hit 2 miss

WAY 0

S0

0x0

blk 0

WAY 1

WAY 2

WAY 3

S1

MISS

0x0

blk 1

S2

S3

U2 - MRU

0 hit 2 miss

WAY 0

S0

0x0

blk 0

WAY 1

WAY 2

WAY 3

S1

MISS

0x0

blk 1

S2

S3

Stats — LRU vs MRU

	LRU	MRU
Total accesses	2	2
Cache hits	0	0
Cache misses	2	2
Hit rate	0.0%	0.0%
Miss rate	100.0%	100.0%
AMAT (cycles)	10	10
Total access time	20	20
Th=1, Tm=10, Tb=1/word · miss penalty = 10 cyc (load-through, block 16w) · AMAT = Th + MissRate(P_miss · Tb)		

m7@cache: ~/trace - serial monitor

2/2 · click to seek

\$ run trace --policy lru,mru

#	blk	set	tag	lru	mr
1	0	s0	0x0	MISS	MISS
2	1	s1	0x0	MISS	MISS

\$